INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA
SALTLAKE
(School of University of Engineering and Management, Kolkata)

Degree: BCA, Stream: BCA
Term – I , Second Semester Examination, February- 2026
Subject Code: BCACC201
Subject Name: Data Structures

Full Marks: 30 | Set No: 2 | Duration: 1 Hour

Part - A
Answer any 5 questions out of 6
Each question carries 2 marks (2 × 5)

1. Explain the importance of data structures in computing. **2**
2. What is the difference between primitive and non-primitive data structures? **2**
3. Define linear and nonlinear data structures with examples. **2**
4. What is the function of a stack pointer? **2**
5. What is an abstract data type (ADT)? **2**
6. What are the advantages of a circular queue over a linear queue? **2**

Part - B
Answer any 2 questions out of 3
Each question carries 5 marks (5 × 2)

7. Write a C function to implement enqueue operation in Circular queue. **5**
8. What do you understand by FIFO principal & How it is different from LIFO? **5**
9. Explain the evaluation of a postfix expression using a stack with an example. **5**

Part - C
Answer any 1 question out of 2
Each question carries 10 marks (10 × 1)

10. Write a C program to implement call by value and call by reference using function. **10**
11. What are the postfix and prefix forms of the expression? **10**
A+B*(C-D)/(P-R)

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